GEOAI 04 change report: Bengaluru (Whitefield): regional check

Totals

Raw detections: 6117
Real changes after filters: 272
Area changed: 68.58 ha
Cropland affected: 14.03 ha (upper bound)
Built-up land affected: 31.3 ha
critical: 1
high: 15
medium: 67
low: 189

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Top 10 changes

#	Tier	Type	Area (ha)	Conf.	Reason
1	critical	built_up_gain	2.875	0.96	new construction / built-up: built-up index fell (-0.20), vegetation index fell
2	high	built_up_gain	1.355	0.971	new construction / built-up: built-up index fell (-0.24), vegetation index fell
3	high	built_up_gain	1.38	0.945	new construction / built-up: water index rose (+0.08), SWIR/blue ratio (paved su
4	high	built_up_gain	1.37	0.945	new construction / built-up: built-up index fell (-0.13), vegetation index fell
5	high	water_loss	0.93	0.985	water receded: built-up index rose (+0.08), vegetation index rose (+0.70), water
6	high	built_up_gain	0.995	0.978	new construction / built-up: built-up index fell (-0.17), vegetation index fell
7	high	built_up_gain	0.83	0.985	new construction / built-up: built-up index fell (-0.20), water index rose (+0.0
8	high	built_up_gain	0.94	0.974	new construction / built-up: built-up index rose (+0.34), vegetation index fell
9	high	built_up_gain	1.385	0.918	new construction / built-up: built-up index fell (-0.13), vegetation index fell
10	high	built_up_gain	1.265	0.925	new construction / built-up: built-up index fell (-0.09), vegetation index fell

Sources and limits

- Copernicus Sentinel-2 L2A (ESA), via the Element84 Earth Search COG bucket
- ESA WorldCover 2021 v200 (10 m), CC-BY 4.0
- Optical imagery cannot see through cloud; flagged areas with cloud are removed, not analysed.
- Change type comes from spectral index rules (no training); check critical changes on the ground.
- 10 m pixels cannot resolve single small buildings or people.
- Rivers, sandbars and lake edges move between dates and can be flagged as change; bare soil in dry regions can look like construction. Check flagged areas near water.
- Land cover is ESA WorldCover 2021, newer than the before image (2019): land already cleared or under construction can be mislabelled as cropland, so treat cropland as an upper bound
- Buildings and road segments need OpenStreetMap, which was not reachable when this result was built: shown as unknown, not zero